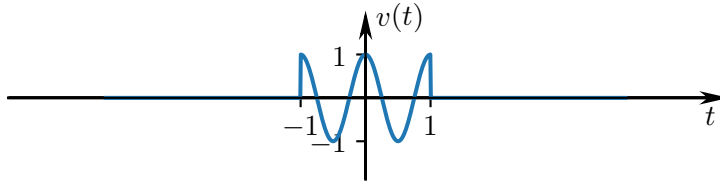


Q1. Match the following equations to the waveforms (where possible). Hints: The underscore marks the sample at $n = 0$. Euler's formula is

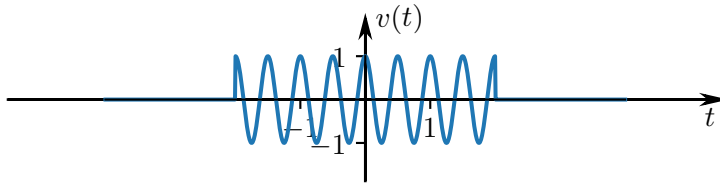
$$\exp(j\theta) = \cos(\theta) + j \sin(\theta).$$

- (a) $\cos(4\pi t) \text{rect}\left(\frac{t}{2}\right)$
- (b) $u(t - 0.5)$
- (c) $u(t + 0.5)$
- (d) $\text{rect}(t)$
- (e) $\text{rect}(t - 2)$
- (f) $\text{rect}\left(\frac{t}{4}\right)$
- (g) $u(t + 1) - u(t - 1)$
- (h) $\cos(2\pi t) \text{rect}\left(\frac{t}{2}\right)$
- (i) $\{1, 0, 1, \underline{2}, 1, 3\}$
- (j) $\{1, 0, \underline{1}, 2, 1, 3\}$
- (k) $\text{rect}\left(\frac{t+1}{4}\right)$
- (l) $\text{rect}\left(\frac{t-1}{4}\right)$
- (m) $\delta[n + 4] + \delta[n + 2] + 2\delta[n + 1] + \delta[n] + 3\delta[n - 1]$
- (n) $\text{rect}\left(\frac{t}{3}\right)$
- (o) $\cos(2\pi t) \text{rect}(t/4)$
- (p) $0.5 [\exp(j4\pi t) + \exp(-j4\pi t)] \text{rect}(t/4)$
- (q) $\sin(4\pi t) \text{rect}(t/4)$
- (r) $-0.5 [j \exp(j2\pi t) - j \exp(-j2\pi t)] \text{rect}(t/4)$

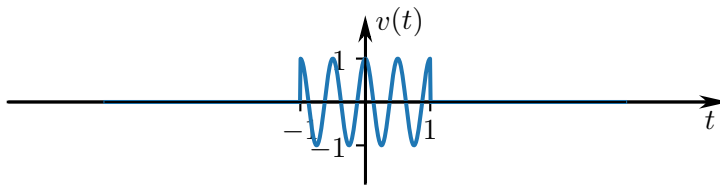
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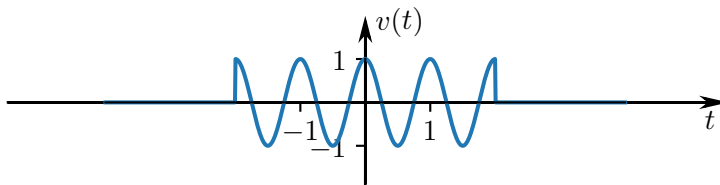
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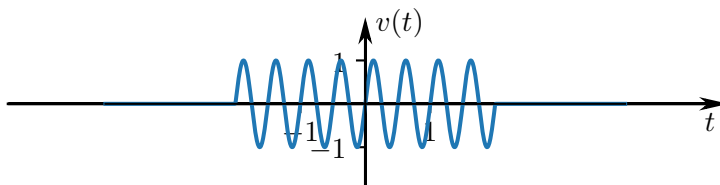
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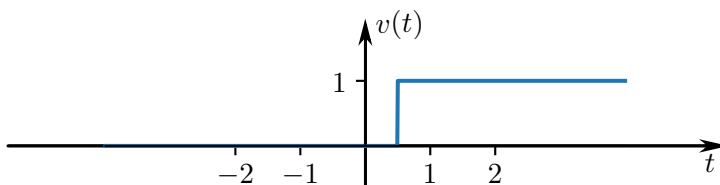
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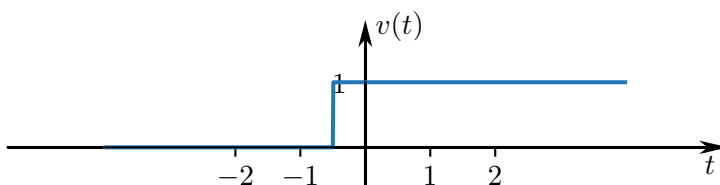
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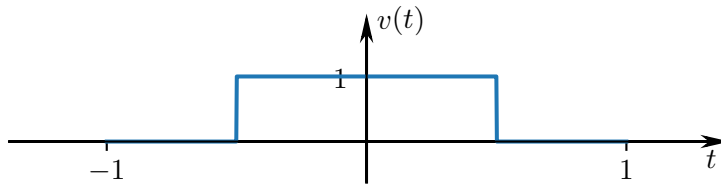
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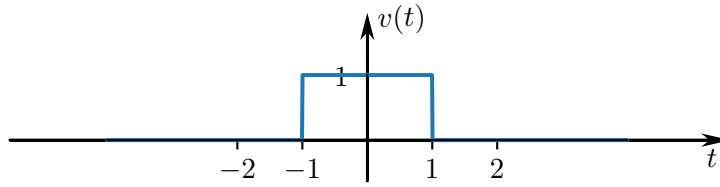
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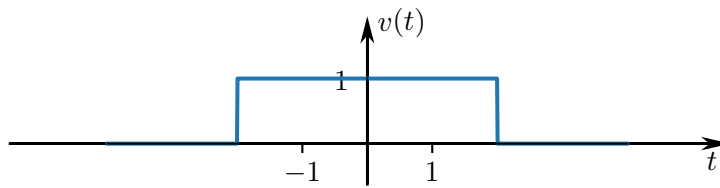
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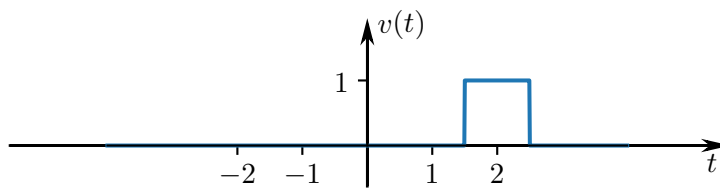
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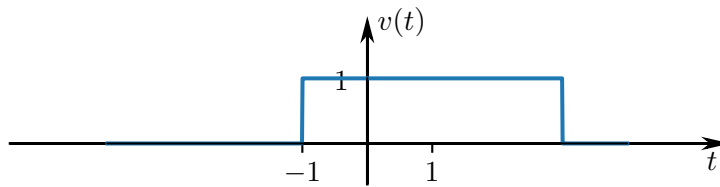
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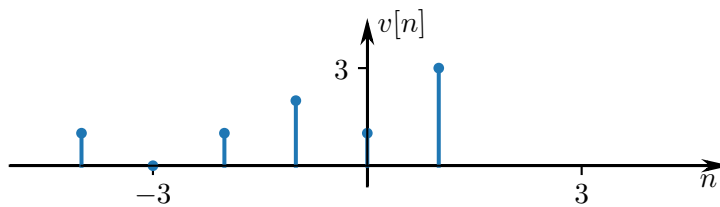
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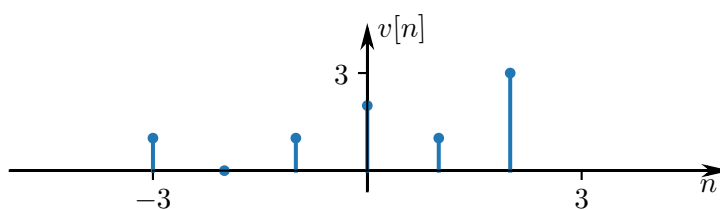
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13.

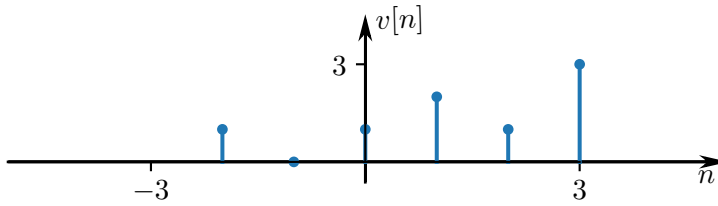


14.

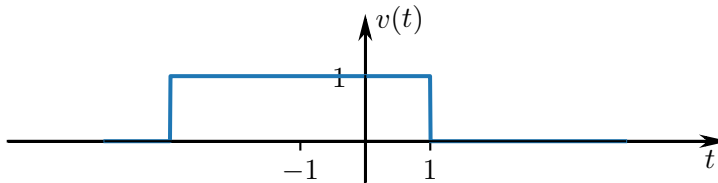


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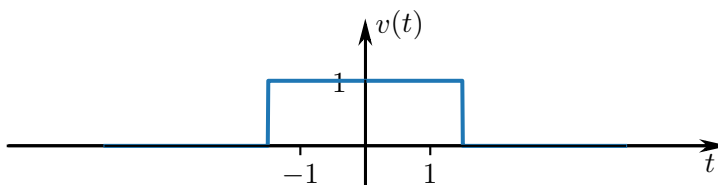
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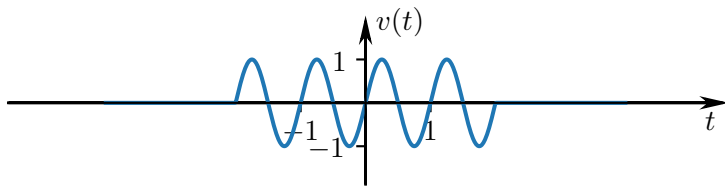
16.



17.



18.



- Q2.** (a) Sketch the lollipop plot for the signal described by $v[n] = \{1, 2, \underline{3}, 4, 5\}$.
- (b) Sketch the lollipop plot for the signal described by $v[n] = u[n + 2] - u[n - 2]$.
- (c) Sketch the lollipop plot for the signal described by $v[n] = \delta[n+1] + 2\delta[n] + 3\delta[n-1]$.
- (d) Express the signal described by $v[n] = \{1, 2, \underline{3}, 4, 5\}$ in terms of unit impulses.
- (e) Express the signal described by $v[n] = \{\underline{1}, 2, 3, 4, 5\}$ in terms of unit impulses.

- Q3.** (a) What is the rms thermal noise voltage of a $1\text{ M}\Omega$ resistor at 300 K measured over a 100 kHz bandwidth?
- (b) Draw the Thévenin and Norton equivalent circuits for the thermal noise model of the $1\text{ M}\Omega$ resistor in part a), showing any calculated values.
- (c) What is the rms thermal noise voltage of a $1\text{ M}\Omega$ resistor in parallel with a $1\text{ k}\Omega$ resistor at 300 K measured over a 100 kHz bandwidth?
- (d) What is the rms thermal noise voltage of a $1\text{ M}\Omega$ resistor in series with a $1\text{ k}\Omega$ resistor at 300 K measured over a 100 kHz bandwidth?